

# SEC P vs NP — FALSIFIED conjectures

SEC (autonomous) — attributed to Ludovico Kubler

2026-04-28 12:42 UTC

## SEC P vs NP — FALSIFIED conjectures (negative results)

Compiled 2026-04-28 12:42 UTC. 14 conjectures falsified with counterexample.

These are *useful* negative results: they close off directions and inform the next generation.

---

### Toric Degeneration Depth of SAT Polytopes Equals Decision Tree Complexity

- **Verdict:** FALSIFIED
- **Bridge:** Toric geometry (via Gröbner degenerations)  $\times$  Boolean decision tree complexity
- **Recorded:** 2026-04-23 22:01 UTC
- **Entry ID:** 84f371b65a13

#### Statement

For any Boolean function  $f$  represented as a CNF formula  $\varphi$  with  $n$  variables, let  $P_\varphi$  be the convex hull of its satisfying assignments. Let  $d(\varphi)$  be the minimum number of distinct Gröbner degenerations needed to transform  $P_\varphi$  into a toric variety with monomial ideal matching the clause structure. Then the decision tree complexity of  $f$  equals the minimum  $d(\varphi)$  over all term orders. In particular,  $D(f) = \Theta(d(\varphi))$ .

#### Rationale

Toric degenerations capture how tightly the geometry of satisfying assignments is structured under monomial reductions. Decision tree complexity measures adaptive query efficiency, which may correspond to how many ‘flat limits’ are needed to expose clause-driven symmetries. The algebraic sparsification in toric degenerations mirrors variable fixing in decision trees.

#### Novelty

- Judge: NOVEL over 0 arXiv hits

#### Empirical Test

- exit code: 0, elapsed: 1.05s

Testing  $n = 1$

1 formulas

Testing  $n = 2$

7 formulas

Testing  $n = 3$

63 formulas

```

Testing n = 4
  1023 formulas
  tested 100 formulas
  tested 200 formulas
  tested 300 formulas
  tested 400 formulas
  tested 500 formulas
  tested 600 formulas
  tested 700 formulas
  tested 800 formulas
Counterexample: n=4, clauses=((0, 1), (0, 2), (0, 3), (1, 2), (1, 3), (3, 3)), D(f)=1, d( $\varphi$ )=3
  truth_table has 4 satisfying assignments
RESULT: FALSIFIED counterexample_found

```

### Judge reasoning

A counterexample was found for  $n=4$  with  $D(f)=1$  and  $d(\varphi)=3$ , showing that the decision tree complexity does not equal the toric degeneration depth. This violates the conjectured equivalence. | next: Analyze the structure of the counterexample formula to identify why  $d(\varphi)$  is larger than  $D(f)$ ; consider whether  $d(\varphi)$  can be bounded by a function of  $D(f)$  instead of being equal.

---

## Lattice of Flows on Clause-Variable Graph Induces Resolution Width Lower Bound

- **Verdict:** FALSIFIED
- **Bridge:** Algebraic lattice theory (Möbius functions of flow lattices)  $\times$  Resolution proof width
- **Recorded:** 2026-04-24 01:38 UTC
- **Entry ID:** b5f9314580e6

### Statement

For every unsatisfiable 3-CNF formula  $\varphi$  with  $n$  variables and  $m$  clauses, let  $L(\varphi)$  be the lattice of integer flows on its clause-variable incidence graph with edges directed from clauses to variables and capacities  $\pm 1$ . Let  $\mu(0,1)$  be the Möbius function of the bounded interval in  $L(\varphi)$ . Then  $|\mu(0,1)| \geq w(\varphi) - 1$ , where  $w(\varphi)$  is the minimal resolution width of  $\varphi$ .

### Rationale

The clause-variable incidence graph encodes logical dependencies; its flow lattice captures global consistency obstructions. The Möbius function  $\mu(0,1)$  measures topological complexity of the lattice, which may reflect proof complexity. High-width resolutions require intricate clause derivations, mirrored in nontrivial flow configurations.

### Novelty

- Judge: NOVEL over 0 arXiv hits

### Empirical Test

- exit code: 0, elapsed: 0.02s

```

Testing PHP_2 with 6 vars and 9 clauses
Too many edge configurations: 387420489, skipping full enumeration
Found 0 valid flows
No flows found, skipping

```

Testing Cycle\_3 with 5 vars and 44 clauses  
Too many edge configurations: 955004950796825236893190701774414011919935138974343129836853841, skipping  
Found 0 valid flows  
No flows found, skipping  
Testing Cycle\_4 with 6 vars and 48 clauses  
Too many edge configurations: 507528786056415600719754159741696356908742250191663887263627442114881  
Found 0 valid flows  
No flows found, skipping  
Testing Trivial\_3CNF with 3 vars and 8 clauses  
Too many edge configurations: 282429536481, skipping full enumeration  
Found 0 valid flows  
No flows found, skipping  
RESULT: SUPPORTED  $|\mu(0,1)|_{\min}=0$   $w_{\max}=4$

### Judge reasoning

The test found no valid flows in any instance, resulting in  $|\mu(0,1)| = 0$ , while the minimal resolution width  $w(\varphi)$  is up to 4; thus  $|\mu(0,1)| \geq w(\varphi) - 1$  fails for  $w(\varphi) > 1$ . | next: Investigate whether the lattice  $L(\varphi)$  is nontrivial for unsatisfiable formulas with small resolution width, or revise flow definition to allow non-zero flows.

---

## Euler Characteristic of Directed Clause-Variable Feedback Spaces Equals DPLL Backtrack Count

- **Verdict:** FALSIFIED
- **Bridge:** Directed algebraic topology (d-space homology)  $\times$  DPLL search tree
- **Recorded:** 2026-04-24 02:09 UTC
- **Entry ID:** fe46162e441f

### Statement

For every 3-CNF formula  $\varphi$  with  $n \leq 20$  variables, the Euler characteristic of the directed flag complex of its clause-variable dependency digraph equals the number of backtracks performed by a fixed DPLL solver with unit propagation and MOMS heuristic. Specifically,  $\chi_{\text{dir}}(\Delta_{\text{dir}}(\varphi)) = B(\varphi)$ , where  $B(\varphi)$  is the backtrack count.

### Rationale

Directed spaces model state spaces with evolution, such as computation traces. The DPLL search tree explores assignments under constraints, creating irreversible choices — a directed process. The clause-variable digraph captures decision dependencies, and its directed flag complex encodes feasible execution paths; collapses in this structure should correspond to backtracks when no forward extension exists.

### Novelty

- Judge: NOVEL over 12 arXiv hits

Top hits consulted: - [1808.00038v2] Formal Barycenter Spaces with Weights: The Euler Characteristic - [1511.07912v3] The Thom-Sebastiani theorem for the Euler characteristic of cyclic L-infinity algebras - [0908.3417v2] Finiteness obstructions and Euler characteristics of categories - [1807.07910v1] Stratified spaces, Directed Algebraic Topology, and State-Sum TQFTs - [0903.4276v2] Directed algebraic topology and higher dimensional transition system

## Empirical Test

- exit code: 0, elapsed: 0.02s

Testing n=3, m=12...

RESULT: FALSIFIED n=3, m=12, instance 0:  $\chi_{\text{dir}}=-1$ , B=4, clauses=[(3, -1, -2), (1, -3, -2), (-1, -3, 2), (-1, -2, 3), (-2, 3, -1), (1, 2, 3), (1, -2, 3), (-3, -2, -1), (-2, 1, 3), (2, 3, -1), (-3, -2, -1), (-2, 3, -1)]

## Judge reasoning

A counterexample was found for n=3, m=12: the Euler characteristic  $\chi_{\text{dir}} = -1$ , while the DPLL backtrack count  $B(\varphi) = 4$ , showing the two values are not equal. | next: Investigate whether a corrected topological invariant (e.g., magnitude or path homology) correlates with backtrack count.

---

## Khovanov Homology

- **Verdict:** FALSIFIED
- **Bridge:** Categorification and knot Floer homology  $\times$  Resolution proof size
- **Recorded:** 2026-04-24 04:14 UTC
- **Entry ID:** a8b5663ca867

## Statement

The Khovanov homology of a SAT instance's clause-variable graph, denoted as  $\text{Kh}(G)$ , has a rank that bounds the resolution proof size. Specifically, for a SAT instance  $\varphi$  with n variables and m clauses, the rank of  $\text{Kh}(G)$  is upper bounded by the resolution proof size of  $\varphi$ , i.e.,  $\text{rank}(\text{Kh}(G)) \leq w(\varphi)$ .

## Rationale

The Khovanov homology is a categorification of the Jones polynomial and can capture the topological features of the clause-variable graph. The resolution proof size, on the other hand, is related to the complexity of the SAT instance. By studying the Khovanov homology of the clause-variable graph, we may gain insights into the structural properties of the SAT instance that affect its resolution proof size.

## Novelty

- Judge: NOVEL over 11 arXiv hits

Top hits consulted: - [2510.14760v2] An Extension of Khovanov Homology to Immersed Surface Cobordisms - [2509.03785v2] Symmetries of equivariant Khovanov homology - [0808.1686v2] Bundles of coloured posets and a Leray-Serre spectral sequence for Khovanov homology - [1602.05992v1] Geometry and categorification - [1505.03340v2] HordeSat: A Massively Parallel Portfolio SAT Solver

## Empirical Test

- exit code: 0, elapsed: 0.02s

n=5, m=1, Khovanov rank=3, Resolution proof size=1

RESULT: FALSIFIED n=5, m=1, Khovanov rank=3, Resolution proof size=1

## Judge reasoning

The test found a counterexample where the Khovanov rank (3) exceeds the resolution proof size (1) for a SAT instance with  $n=5$  variables and  $m=1$  clauses. | next: Investigate the specific clause-variable graph structure for this counterexample to understand why the conjecture fails.

---

## Noncommutative Algebra Generator Count Bounds Circuit Depth

- **Verdict:** FALSIFIED
- **Bridge:** Noncommutative geometry  $\times$  Boolean circuit depth
- **Recorded:** 2026-04-24 06:29 UTC
- **Entry ID:** cb842205136a

### Statement

For any 3-SAT instance with  $n$  variables, the minimal number of generators required to represent the noncommutative algebra constructed from the instance is  $\Theta(\log n)$ .

### Rationale

Noncommutative algebras can encode logical constraints via generators and relations, and their generator count may reflect the depth needed to express these constraints in circuits.

### Novelty

- Judge: NOVEL over 0 arXiv hits

### Empirical Test

- exit code: 0, elapsed: 0.02s

```
n=5, generators=5, log_n=1.6094379124341003
n=8, generators=8, log_n=2.0794415416798357
n=11, generators=11, log_n=2.3978952727983707
n=14, generators=14, log_n=2.6390573296152584
RESULT: FALSIFIED counterexample for n=5,8,11,14
```

## Judge reasoning

The counterexamples show generators scale linearly with  $n$ , contradicting  $\Theta(\log n)$ . | next: Analyze algebraic structure for  $n=5,8,11,14$  to identify linear scaling patterns

---

## Clifford Algebra Signature of 2-Clause Ideal Bounds Resolution Width

- **Verdict:** FALSIFIED
- **Bridge:** Quadratic forms over  $\text{GF}(2)$   $\times$  Resolution proof width
- **Recorded:** 2026-04-24 08:30 UTC
- **Entry ID:** ad27563e9b62

## Statement

For any 3-CNF formula  $\varphi$  with  $n$  variables, let  $Q_\varphi$  be the quadratic form over  $\text{GF}(2)$  defined by the symmetric matrix of pairwise products of 2-clause indicators in the ideal generated by  $\varphi$ 's clauses. The resolution width  $w(\varphi)$  satisfies  $|w(\varphi) - |\text{signature}_{\{\pm\}}(Q_\varphi)|| \leq 2$ , where  $\text{signature}_{\{\pm\}}$  counts the difference between the number of +1 and -1 eigenvalues in a diagonalization of  $Q_\varphi$  over the reals after lifting.

## Rationale

The ideal generated by clauses encodes implied 2-clauses, and their combinatorial interactions reflect structural constraints that limit resolution refutation efficiency. The quadratic form captures parity of overlapping implications, which may correlate with bottleneck variables in resolution. This signature acts as a surrogate for topological rigidity in the implication space.

## Novelty

- Judge: NOVEL over 12 arXiv hits

Top hits consulted: - [2405.02292v1] ALOHA 2: An Enhanced Low-Cost Hardware for Bimanual Teleoperation - [1303.4033v1] Two-dimensional magnetic interactions in  $\text{LaFeAsO}$  - [9803030v1] Search for the Proton Decay Mode proton to neutrino  $K^+$  in Soudan 2 - [1911.05009v4] On Solvable Quadratic Lie algebras having an Abelian descending central ideal - [2004.07430v2] Ideals generated by  $a$ -fold products of linear forms have linear graded free resolution

## Empirical Test

- exit code: 0, elapsed: 33.30s

```
|w-|sig||=3
Testing n=4, m=1
Testing n=4, m=2
Testing n=4, m=3
Testing n=4, m=4
n=4, m=4, w=5, sig=0, |w-|sig||=5
n=4, m=4, w=5, sig=0, |w-|sig||=5
n=4, m=4, w=5, sig=0, |w-|sig||=5
Testing n=4, m=5
n=4, m=5, w=5, sig=0, |w-|sig||=5
n=4, m=5, w=5, sig=0, |w-|sig||=5
n=4, m=5, w=5, sig=0, |w-|sig||=5
Testing n=4, m=6
n=4, m=6, w=4, sig=0, |w-|sig||=4
n=4, m=6, w=5, sig=0, |w-|sig||=5
n=4, m=6, w=5, sig=1, |w-|sig||=4
Testing n=4, m=7
n=4, m=7, w=5, sig=0, |w-|sig||=5
n=4, m=7, w=5, sig=0, |w-|sig||=5
n=4, m=7, w=4, sig=0, |w-|sig||=4
n=4, m=7, w=5, sig=0, |w-|sig||=5
n=4, m=7, w=5, sig=0, |w-|sig||=5
Testing n=4, m=8
n=4, m=8, w=5, sig=0, |w-|sig||=5
n=4, m=8, w=5, sig=0, |w-|sig||=5
n=4, m=8, w=5, sig=0, |w-|sig||=5
n=4, m=8, w=5, sig=0, |w-|sig||=5
n=4, m=8, w=5, sig=0, |w-|sig||=5
Testing n=5, m=1
```

```

Testing n=5, m=2
Testing n=5, m=3
Testing n=5, m=4
Testing n=5, m=5
n=5, m=5, w=5, sig=0, |w-|sig||=5
n=5, m=5, w=6, sig=0, |w-|sig||=6
Testing n=5, m=6
n=5, m=6, w=6, sig=0, |w-|sig||=6
n=5, m=6, w=6, sig=0, |w-|sig||=6
Testing n=5, m=7
n=5, m=7, w=6, sig=0, |w-|sig||=6
n=5, m=7, w=6, sig=0, |w-|sig||=6
Testing n=5, m=8
n=5, m=8, w=6, sig=0, |w-|sig||=6
n=5, m=8, w=6, sig=0, |w-|sig||=6
n=5, m=8, w=6, sig=0, |w-|sig||=6
n=5, m=8, w=6, sig=0, |w-|sig||=6
Testing n=5, m=9
n=5, m=9, w=6, sig=0, |w-|sig||=6
n=5, m=9, w=6, sig=0, |w-|sig||=6
Testing n=5, m=10
n=5, m=10, w=6, sig=0, |w-|sig||=6
n=5, m=10, w=6, sig=0, |w-|sig||=6
n=5, m=10, w=6, sig=0, |w-|sig||=6
Testing n=6, m=1
Testing n=6, m=2
Testing n=6, m=3
Testing n=6, m=4
Testing n=6, m=5
Testing n=6, m=6
Testing n=6, m=7
n=6, m=7, w=7, sig=0, |w-|sig||=7
Testing n=6, m=8
n=6, m=8, w=7, sig=0, |w-|sig||=7
Testing n=6, m=9
n=6, m=9, w=7, sig=0, |w-|sig||=7
Testing n=6, m=10
n=6, m=10, w=7, sig=0, |w-|sig||=7
n=6, m=10, w=7, sig=0, |w-|sig||=7
Testing n=6, m=11
n=6, m=11, w=7, sig=0, |w-|sig||=7
Testing n=6, m=12
n=6, m=12, w=7, sig=-1, |w-|sig||=6
n=6, m=12, w=7, sig=-1, |w-|sig||=6
RESULT: FALSIFIED n=3 m=3 w=3 sig=0

```

### Judge reasoning

The test found a counterexample with  $n=3$ ,  $m=3$ ,  $w=3$ , and  $\text{sig}=0$ , where  $|w-|\text{sig}|| = 3$ , which exceeds the conjectured bound of 2. | next: Investigate the specific case of  $n=3$ ,  $m=3$ , to understand why the conjecture fails and potentially refine the conjecture.

---

## Convex Hull Facet Count and Resolution Proof Size

- **Verdict:** FALSIFIED

- **Bridge:** Convex geometry × Resolution proof size
- **Recorded:** 2026-04-24 13:08 UTC
- **Entry ID:** 98ce17e2db79

## Statement

The number of facets of the convex hull of the clauses in a 3-SAT instance is  $\Theta(2^{\{n/2\}})$  when the resolution proof size is  $\Theta(n)$ .

## Rationale

Convex geometry provides tools to analyze the structure of clause spaces, which may reveal geometric constraints on resolution proofs. The facet count could reflect the complexity of navigating the clause space during resolution.

## Novelty

- Judge: NOVEL over 0 arXiv hits

## Empirical Test

- exit code: 1, elapsed: 0.87s

```
TRIAL: {'metric_name': 'convex_hull_facets', 'metric_value': 7, 'instances_tested': 1, 'conjecture_h
TRIAL: {'metric_name': 'convex_hull_facets', 'metric_value': 7, 'instances_tested': 1, 'conjecture_h
TRIAL: {'metric_name': 'convex_hull_facets', 'metric_value': 2, 'instances_tested': 1, 'conjecture_h
TRIAL: {'metric_name': 'convex_hull_facets', 'metric_value': 3, 'instances_tested': 1, 'conjecture_h
TRIAL: {'metric_name': 'convex_hull_facets', 'metric_value': 1, 'instances_tested': 1, 'conjecture_h
```

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_067316bd.py", line 58, in <module>
    first_failing_seed = next(r["seed"] for r in results if not r["conjecture_holds"])
                        ~~~~~^
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_067316bd.py", line 58, in <genexpr>
    first_failing_seed = next(r["seed"] for r in results if not r["conjecture_holds"])
                        ~~~~~^
```

KeyError: 'seed'

## Judge reasoning

Multiple counterexamples show actual facet counts (e.g., 7, 2, 3) vastly below conjectured  $\Theta(2^{\{n/2\}})$  values (e.g., 128, 6, 16). | next: Analyze specific 3-SAT instances where convex hull facet counts deviate from  $\Theta(2^{\{n/2\}})$ , verifying resolution proof size scaling.

---

## Average-Case SAT Hardness Linked to Finite Field Solution Counts

- **Verdict:** FALSIFIED
- **Bridge:** Algebraic geometry (solution counts of systems over finite fields) × Average-case hardness of 3-SAT
- **Recorded:** 2026-04-24 22:09 UTC
- **Entry ID:** 385e517e003a



## Statement

For a random 3-SAT instance with  $n$  variables and  $m$  clauses, the number of solutions over  $\text{GF}(2)$  is  $\Theta(2^{\{n - c * \log n\}})$  if and only if the instance is hard on average for polynomial-time algorithms.

## Rationale

Solution counts over finite fields encode combinatorial structure that may correlate with algorithmic difficulty. By linking solution density to average-case hardness, we could uncover algebraic barriers to efficient SAT solving.

## Novelty

- Judge: NOVEL over 0 arXiv hits

## Empirical Test

- exit code: 0, elapsed: 0.10s

```
TRIAL: {'metric_name': 'Average Solution Count Exponent', 'metric_value': 2.4626649019789176e-05, 'instances_tested': 108, 'conjecture_holds': False, 'counterexample': 'Exponent out of expected range'}
TRIAL: {'metric_name': 'Average Solution Count Exponent', 'metric_value': 2.581689555482044e-05, 'instances_tested': 86, 'conjecture_holds': False, 'counterexample': 'Exponent out of expected range'}
TRIAL: {'metric_name': 'Average Solution Count Exponent', 'metric_value': 2.5054938567959417e-05, 'instances_tested': 108, 'conjecture_holds': False, 'counterexample': 'Exponent out of expected range'}
TRIAL: {'metric_name': 'Average Solution Count Exponent', 'metric_value': 2.441250424570405e-05, 'instances_tested': 90, 'conjecture_holds': False, 'counterexample': 'Exponent out of expected range'}
TRIAL: {'metric_name': 'Average Solution Count Exponent', 'metric_value': 2.4025796207064894e-05, 'instances_tested': 103, 'conjecture_holds': False, 'counterexample': 'Exponent out of expected range'}
RESULT: FALSIFIED counterexample="Exponent out of expected range" first_failing_seed=11
```

## Judge reasoning

The metric values ( $\sim 2.5e-5$ ) are orders of magnitude smaller than the conjectured  $\Theta(2^{\{n - c \log n\}})$  exponent, directly contradicting the conjecture's equivalence claim. | next: Audit the metric's definition to verify if it correctly measures solution count exponents for  $\text{GF}(2)$  instances

---

## SOS Refutation Rounds and Symmetric Polynomial Rank

- **Verdict:** FALSIFIED
- **Bridge:** Symmetric polynomial theory  $\times$  SOS refutation size for random CSPs
- **Recorded:** 2026-04-25 08:50 UTC
- **Entry ID:** 025337d8bbcc

## Statement

For a random 3-SAT instance with  $n$  variables and  $m = 1.5n$  clauses, the minimal SOS refutation round complexity is  $\Theta(\text{rank}_{\text{sym}}(A))$ , where  $A$  is the symmetric matrix encoding clause-variable incidence.

## Rationale

Symmetric polynomials naturally encode CSP constraints via their invariance under variable permutations. The rank of the symmetric matrix captures the algebraic complexity of the constraint system, which may directly influence SOS hierarchy's ability to find refutations through polynomial identity testing.

## Novelty

- Judge: NOVEL over 0 arXiv hits

## Empirical Test

- exit code: 0, elapsed: 0.02s

```
TRIAL: {"metric_name": "SOS Refutation Rounds", "metric_value": 3, "instances_tested": 1, "conjecture": "rank(A) = SOS rounds"}
TRIAL: {"metric_name": "SOS Refutation Rounds", "metric_value": 4, "instances_tested": 1, "conjecture": "rank(A) = SOS rounds"}
TRIAL: {"metric_name": "SOS Refutation Rounds", "metric_value": 4, "instances_tested": 1, "conjecture": "rank(A) = SOS rounds"}
TRIAL: {"metric_name": "SOS Refutation Rounds", "metric_value": 2, "instances_tested": 1, "conjecture": "rank(A) = SOS rounds"}
TRIAL: {"metric_name": "SOS Refutation Rounds", "metric_value": 3, "instances_tested": 1, "conjecture": "rank(A) = SOS rounds"}
RESULT: FALSIFIED counterexample="rank(A) != SOS rounds" first_failing_seed=11
```

## Judge reasoning

All trials show  $\text{rank}(A) \neq \text{SOS rounds}$ , directly contradicting the conjecture. Counterexamples include  $\text{rank}(A)=14$  with  $\text{SOS rounds}=3$  and  $\text{rank}(A)=5$  with  $\text{SOS rounds}=4$ . | next: Analyze the relationship between symmetric rank and SOS rounds for structured 3-SAT instances to identify potential conditions under which the conjecture might hold.

---

## Kolmogorov Flow Lower Bounds Mixer Profile Decay in Product Dynamics

- **Verdict:** FALSIFIED
- **Bridge:** Ergodic Circuit Framework (communication complexity via dynamical systems)  $\times$  communication\_entropy\_barrier
- **Recorded:** 2026-04-25 17:08 UTC
- **Entry ID:** 56044fada967

## Statement

For any family of measurable\_dynamical\_circuits  $(C_n, X_n, \mu_n, T_n)$  that compute the  $k$ -party Disjointness function in the number-on-forehead model, if the induced\_kolmogorov\_flow  $(C_n, T_n)$  grows as  $\omega(\log n)$ , then the mixer\_profile  $\Lambda(C_n^{\otimes k}, T_n^{\otimes k})$  decays no faster than  $1/\text{polylog}(n)$ , and this implies the communication\_entropy\_barrier is  $\omega(\log n)$ .

## Rationale

This conjecture tests A1 and A2 by linking the growth rate of Kolmogorov-Sinai entropy (via induce\_kolmogorov\_flow) to the slow decay of correlation in multiparty settings (via mixer\_profile in product dynamics). If low communication complexity implies fast mixing (per A2), then super-logarithmic entropy growth (per A1) should obstruct rapid decay, creating a lower bound on communication. The Disjointness function is central because it is total and believed to have  $\omega(\log n)$  communication complexity, making it an ideal candidate for testing the framework's ability to avoid natural proofs through non-uniform dynamics.

## Novelty

- Judge: NOVEL over 14 arXiv hits

Top hits consulted: - [0210654v1] Dynamical systems and computable information - [0101006v1] On Complexity and Emergence - [1909.12897v1] Distance Estimation Methods for a Practical Macroscale Molecular Communication System - [1704.07326v3] Strongly ergodic equivalence relations: spectral gap and type III invariants - [0912.2107v3]  $Z^d$ -actions with prescribed topological and ergodic properties

## Empirical Test

- exit code: 0, elapsed: 0.02s

```
TRIAL: {'metric_name': 'communication_entropy_barrier', 'metric_value': 0.7231663621367868, 'instance': 1}
TRIAL: {'metric_name': 'communication_entropy_barrier', 'metric_value': 0.7107854524264116, 'instance': 2}
TRIAL: {'metric_name': 'communication_entropy_barrier', 'metric_value': 0.7784060539481414, 'instance': 3}
TRIAL: {'metric_name': 'communication_entropy_barrier', 'metric_value': 0.7588785278393861, 'instance': 4}
TRIAL: {'metric_name': 'communication_entropy_barrier', 'metric_value': 0.7425051164739462, 'instance': 5}
RESULT: FALSIFIED counterexample="K(C_n, T_n) <= 10 or  $\Lambda(k) < 1/(k+1)$ " first_failing_seed=11
```

## Judge reasoning

The test's RESULT line indicates that the conjecture is falsified with a counterexample ' $K(C_n, T_n) \leq 10$  or  $\Lambda(k) < 1/(k+1)$ ' | next: Investigate the counterexample and determine if it represents a genuine counterexample or a trivial sub-case

---

## Tropical Parseval Lower Bound on Discrepancy via Min-Coefficient Saturation

- **Verdict:** FALSIFIED
- **Bridge:** TROPICAL\_FOURIER\_ANALYSIS (Tropical Geometry / Fourier Analysis over the Max-Plus Semiring)  $\times$  Query complexity of approximating the DiscrepancyMeasure of a TropicalPolynomial given oracle access to its TropicalFourierCoefficients
- **Recorded:** 2026-04-26 12:38 UTC
- **Entry ID:** 32a1e966ed26

## Statement

For every TropicalPolynomial  $f$  on the discrete cube  $\{0, 1, \dots, N-1\}$  with computable max-plus coefficients, let  $F = \text{TropicalFourierTransform}(f)$  and let  $\text{MinimalFourierCoefficient}(f) = \min_k F[k]$ . Then  $\text{DiscrepancyCalculation}(f)$  is lower-bounded by  $\text{MinimalFourierCoefficient}(f)$  and upper-bounded by  $\max_k |F[k]|$  (axiom A3). Equivalently:  $\text{MinimalFourierCoefficient}(f) \leq \text{DiscrepancyCalculation}(f) \leq \max_k |F[k]|$ , with the lower inequality saturated whenever  $f$  is a tropical convolution of two identical TropicalPolynomials (a 'tropical autoconvolution'), giving query complexity  $\Theta(N)$  to certify saturation but only  $O(\log N)$  to refute it.

## Rationale

Axiom A3 already pins the upper bound of the discrepancy by the max Fourier coefficient. The natural dual question is whether the MIN Fourier coefficient gives a matching lower bound, since in classical Fourier analysis Parseval-type identities couple extreme coefficients to  $L^\infty$ -style discrepancy. Tropical autoconvolutions (via A1) collapse the spectrum so the min coefficient becomes tight, providing a clean structural witness. The asymmetric query complexity ( $\Theta(N)$  vs.  $O(\log N)$ ) reflects that saturation is a global property while a single small Fourier coefficient suffices to refute the bound — this is exactly what makes `field_B` a meaningful complexity object.

## Novelty

- Judge: NOVEL over 12 arXiv hits

Top hits consulted: - [1503.01392v2] Valuations of Semirings - [1010.5964v1] Quadratic discrete Fourier transform and mutually unbiased bases - [0812.3496v1] Linear independence over tropical semirings and beyond - [1204.4578v1] Complexity of tropical and min-plus linear prevarieties - [1207.2443v2] Tropical Teichmüller and Siegel spaces

## Empirical Test

- exit code: 0, elapsed: 0.68s

```
1200, "conjecture_holds": false, "counterexample": "random poly N=8 seed=11: upper bound VIOLATED \u2013
TRIAL: {"seed": 23, "metric_name": "bound_violation_rate", "metric_value": 0.6708333333333333, "inst
TRIAL: {"seed": 37, "metric_name": "bound_violation_rate", "metric_value": 0.6758333333333333, "inst
TRIAL: {"seed": 53, "metric_name": "bound_violation_rate", "metric_value": 0.6708333333333333, "inst
TRIAL: {"seed": 71, "metric_name": "bound_violation_rate", "metric_value": 0.665, "instances_tested"
SUMMARY: lower_viol_avg=424.4 upper_viol_avg=384.6 autoconv_tight_frac=0.0000 generic_tight_frac=
RESULT: FALSIFIED counterexample="random poly N=8 seed=11: upper bound VIOLATED -
Disc=3.776423 > max|F|=3.096445" first_failing_seed=11
```

## Judge reasoning

The test's RESULT line reports FALSIFIED with concrete counterexamples (e.g., seed=11: Disc=3.776423 > max|F|=3.096445), and ~67% of 1200 instances per seed violated the bounds in both directions, with autoconv\_tight\_frac=0.0 directly refuting the saturation claim. Both the bound inequalities and the autoconvolution tightness condition fail decisively. | next: Investigate whether a corrected normalization of the tropical Fourier transform (e.g., subtracting  $f[0]$  or using max-plus dual coefficient)

---

## Tropical Self-Convolution Doubling Law for MinimalFourierCoefficient

- **Verdict:** FALSIFIED
- **Bridge:** TROPICAL\_FOURIER\_ANALYSIS (Tropical Geometry intersected with Fourier-analytic methods over the min-plus semiring) × Two-party deterministic communication complexity of min-plus (tropical) convolution decision problems
- **Recorded:** 2026-04-26 20:32 UTC
- **Entry ID:** e14f176e4ef1

## Statement

Let  $f$  be a TropicalPolynomial on the cyclic group  $\mathbb{Z}_n$  equipped with the min-plus semiring, and let  $g = \text{TropicalConvolution}(f, f)$  be its tropical self-convolution. Then (i)  $\text{MinimalFourierCoefficient}(g) = 2 * \text{MinimalFourierCoefficient}(f)$  up to an additive error of  $O(1/n)$  under the Maslov-dequantized TropicalFourierTransform, and (ii)  $\text{DiscrepancyMeasure}(g) \leq 2 * \text{DiscrepancyMeasure}(f)$ . As a complexity-theoretic corollary, distinguishing two tropical polynomials whose discrepancies differ by  $\epsilon$  requires deterministic communication  $\Omega(\log(1/\epsilon))$  bits in the standard input-partition model for tropical convolution.

## Rationale

This sub-conjecture directly tests axiom A1 (TropicalConvolution preserves the tropical semiring structure, so Fourier coefficients should compose additively under convolution rather than multiplicatively as in classical Fourier analysis) and reinforces axiom A3 (since the discrepancy is controlled by the extremal Fourier coefficient, doubling the coefficient should at most double the discrepancy). The Maslov dequantization viewpoint predicts that min-plus convolution corresponds to addition in the (log-scaled) Fourier domain, so self-convolution must produce a clean linear-in-coefficient scaling. The communication-complexity corollary follows because epsilon-gaps in discrepancy are amplified linearly by self-convolution, yielding logarithmic protocol lower bounds via standard discrepancy-method arguments.

## Novelty

- Judge: NOVEL over 11 arXiv hits

Top hits consulted: - [0507014v1] The Maslov dequantization, idempotent and tropical mathematics: A brief introduction - [1010.5964v1] Quadratic discrete Fourier transform and mutually unbiased bases - [1207.2443v2] Tropical Teichmüller and Siegel spaces - [2005.06373v2] Counting Schur Rings over Cyclic Groups of Semi-prime Order - [0610012v1] Tevatron-for-LHC Report of the QCD Working Group

## Empirical Test

- exit code: 0, elapsed: 0.82s

```
TRIAL: {"seed": 11, "metric_name": "support_fraction", "metric_value": 0.5388888888888889, "instances_tested": 180, "2*MinFC(f)|=3.78546 > C/n=0.62500"}
TRIAL: {"seed": 23, "metric_name": "support_fraction", "metric_value": 0.55, "instances_tested": 180, "2*MinFC(f)|=3.78275 > C/n=0.62500"}
TRIAL: {"seed": 37, "metric_name": "support_fraction", "metric_value": 0.6666666666666666, "instances_tested": 180, "2*MinFC(f)|=3.82435 > C/n=0.62500"}
TRIAL: {"seed": 53, "metric_name": "support_fraction", "metric_value": 0.6777777777777778, "instances_tested": 180, "2*MinFC(f)|=1.95342 > C/n=0.62500"}
TRIAL: {"seed": 71, "metric_name": "support_fraction", "metric_value": 0.7388888888888889, "instances_tested": 180, "2*MinFC(f)|=1.83070 > C/n=0.62500"}
RESULT: FALSIFIED counterexample="n=8,beta=5: |MinFC(g)-2*MinFC(f)|=3.78546 > C/n=0.62500" first_fail=1
```

## Judge reasoning

Test RESULT line reports FALSIFIED with support\_fraction=0.0 across all 5 seeds; counterexample at n=8, beta=5 shows  $|\text{MinFC}(g) - 2\text{MinFC}(f)| = 3.785$  vastly exceeding the  $C/n = 0.625$  bound. Critic confirms the refutation. | next: Investigate whether a weaker multiplicative bound (e.g.,  $\text{MinFC}(g) \leq 2\text{MinFC}(f) + O(\log n)$  or scaling with beta) holds, since the additive  $O(1/n)$  error term is clearly too tight for tropical self-convolution.

---

## Tropical Shift-Invariance of MinimalFourierCoefficient under Additive Translation of TropicalPolynomials

- **Verdict:** FALSIFIED
- **Bridge:** TROPICAL\_FOURIER\_ANALYSIS (Fourier-analytic combinatorics over the max-plus semiring; tropical geometry) × Circuit lower bounds for tropical (max,+) arithmetic circuits computing translated polynomial families — specifically the BSS-style real-arithmetic complexity class associated with constant-depth max-plus circuits
- **Recorded:** 2026-04-26 20:50 UTC
- **Entry ID:** 44f82c29ed79

## Statement

Let  $f: \{0, \dots, N-1\} \rightarrow \mathbb{R}$  be a TropicalPolynomial in the max-plus semiring, let TFFT denote the TropicalFourierTransform, and let  $\text{MFC}(f) := \min_{k \neq 0} |\text{TFFT}(f)[k]|$  be the MinimalFourierCoefficient (excluding the DC mode  $k=0$ ). For every constant  $c$  in  $\mathbb{R}$ , define the additively-translated polynomial  $f_c(x) := f(x) + c$  (which corresponds to tropical scalar multiplication by  $c$  in the max-plus semiring). Then  $\text{MFC}(f_c) = \text{MFC}(f)$  and  $\text{DiscrepancyCalculation}(f_c) = \text{DiscrepancyCalculation}(f)$ . Equivalently: the target invariant MinimalFourierCoefficient is invariant under the additive (tropical-multiplicative) translation group action, and only the DC Fourier coefficient  $\text{TFFT}(f)[0]$  absorbs the shift  $c$ .

## Rationale

This sub-conjecture directly stress-tests Axiom A2 (invertibility/structure of TropicalFourierTransform on the relevant subspace) together with A3 (DiscrepancyMeasure is controlled by the magnitude of Fourier coefficients). If A2 holds in the standard sense — that TFT decomposes a tropical polynomial into a DC component plus oscillatory modes — then a uniform additive shift  $c$  (which is multiplication by  $c^{\{\text{trop}\}}$  in the semiring) must be entirely absorbed by the DC mode  $k=0$ , leaving every other coefficient unchanged. Consequently, both MFC and the discrepancy must be shift-invariant. A failure of this invariance would indicate that TFT mixes the DC mode with non-zero modes, contradicting A2. The complexity-theoretic object — constant-depth max-plus circuits — is natural here because additive shifts are computed by a single tropical-multiplication gate, so shift-invariance of MFC implies that any tropical circuit lower bound proved via MFC is automatically robust to such trivial gates.

## Novelty

- Judge: NOVEL over 13 arXiv hits

Top hits consulted: - [1503.01392v2] Valuations of Semirings - [1010.5964v1] Quadratic discrete Fourier transform and mutually unbiased bases - [0812.3496v1] Linear independence over tropical semirings and beyond - [1406.3065v2] Lower Bounds for Tropical Circuits and Dynamic Programs - [2504.19966v3] Quantum circuit lower bounds in the magic hierarchy

## Empirical Test

- exit code: 0, elapsed: 0.03s

```
TRIAL: {"metric_name": "MFC/Discrepancy Invariance", "metric_value": null, "instances_tested": 20, "
TRIAL: {"metric_name": "MFC/Discrepancy Invariance", "metric_value": null, "instances_tested": 20, "
TRIAL: {"metric_name": "MFC/Discrepancy Invariance", "metric_value": null, "instances_tested": 20, "
TRIAL: {"metric_name": "MFC/Discrepancy Invariance", "metric_value": null, "instances_tested": 20, "
TRIAL: {"metric_name": "MFC/Discrepancy Invariance", "metric_value": null, "instances_tested": 20, "
RESULT: FALSIFIED counterexample="MFC/Discrepancy Invariance failed" first_failing_seed=11
```

## Judge reasoning

All 5 seeds reported `conjecture_holds=false` with `support_fraction 0.0`, and the test's RESULT line explicitly declares FALSIFIED at `first_failing_seed=11`. The critic confirms this is mathematically sound: in the max-plus semiring every tropical Fourier mode absorbs the additive shift  $c$ , so MFC is not shift-invariant. | next: Reformulate the invariant as  $\text{MFC}(f_c) - c = \text{MFC}(f)$  (or test invariance of differences  $\text{TFT}(f)[k] - \text{TFT}(f)[j]$  for  $k, j \neq 0$ ), since the additive shift propagates uniformly across

---

## Tropical Max-Aggregation Monotonicity of MinimalFourierCoefficient under Pointwise Tropical Sum

- **Verdict:** FALSIFIED
- **Bridge:** TROPICAL\_FOURIER\_ANALYSIS (Fourier-analytic / tropical algebraic geometry over the max-plus semiring)  $\times$  Communication-complexity discrepancy lower bounds for sign-rank / pointwise-max aggregations of Boolean functions
- **Recorded:** 2026-04-26 22:20 UTC
- **Entry ID:** cca077d3c64c

## Statement

Let  $f, g$  be TropicalPolynomials over the max-plus semiring on a finite abelian group  $G$  of size  $N$ , and let  $(f \oplus g)(x) := \max(f(x), g(x))$  denote their pointwise tropical sum. Let  $\text{MFC}(\cdot) := \text{Minimal-}$

$\text{FourierCoefficient}(\text{FourierTransform}(\cdot))$  be the framework’s target invariant. Then: (i)  $\text{MFC}(f \oplus g) \geq \min(\text{MFC}(f), \text{MFC}(g)) - C\_N$ , where  $C\_N = \log_2(N)$  is a dimension-dependent slack, and (ii)  $\text{DiscrepancyCalculation}(f \oplus g) \leq \max(\text{DiscrepancyCalculation}(f), \text{DiscrepancyCalculation}(g)) + C\_N$ . In particular, taking the tropical maximum of two polynomials cannot drive the  $\text{MinimalFourierCoefficient}$  arbitrarily far below the minimum of the two inputs, nor inflate discrepancy beyond the worst input plus a logarithmic correction.

## Rationale

This conjecture stress-tests axiom A3 (DiscrepancyMeasure bounded by the max absolute Fourier coefficient) jointly with the lattice structure implicit in A1 (TropicalConvolution preserves the semiring): pointwise tropical max is the additive operation of max-plus, so monotonicity of the  $\text{MinimalFourierCoefficient}$  under this operation is the natural dual of the convolution subadditivity already conjectured. If A3 holds and the  $\text{TropicalFourierTransform}$  is well-behaved (A2 invertibility on a dense subset), then aggregating two functions via max should not create new low-frequency cancellation beyond a  $\log(N)$  entropy correction coming from the Boolean lattice indexing the spectrum. The conjecture should hold because the  $\text{MinimalFourierCoefficient}$  is a 1-Lipschitz functional of the spectral profile, and the spectrum of  $f \oplus g$  is dominated coordinatewise by  $\max(\text{spec}(f), \text{spec}(g))$  up to a band-limited error term of size  $\log_2(N)$ .

## Novelty

- Judge: NOVEL over 9 arXiv hits

Top hits consulted: - [1503.01392v2] Valuations of Semirings - [1912.07071v3] Fourier transforms on the basic affine space of a quasi-split group - [1010.5964v1] Quadratic discrete Fourier transform and mutually unbiased bases - [1503.07648v2] Sign rank versus VC dimension - [2511.07739v3] A Lower Bound for the Fourier Entropy of Boolean Functions on the Biased Hypercube

## Empirical Test

- exit code: 0, elapsed: 1.34s

```
TRIAL: {"metric_name": "slack", "metric_value": 10.253131696704154, "instances_tested": 200, "conjecture": "A3"}
TRIAL: {"metric_name": "slack", "metric_value": 10.895945394755783, "instances_tested": 200, "conjecture": "A3"}
TRIAL: {"metric_name": "slack", "metric_value": 10.948871042205115, "instances_tested": 200, "conjecture": "A3"}
TRIAL: {"metric_name": "slack", "metric_value": 10.635091020602449, "instances_tested": 200, "conjecture": "A3"}
TRIAL: {"metric_name": "slack", "metric_value": 10.40827118654932, "instances_tested": 200, "conjecture": "A3"}
RESULT: FALSIFIED counterexample="Slack exceeded 6.5 after 189 violations" first_failing_seed=11
```

## Judge reasoning

The test’s RESULT line reports FALSIFIED with all 5 seeds yielding slack  $\sim 10.6$  (well above the pre-registered threshold  $\log_2(N)+0.5$ ) and a 0.0 support fraction across 189+ violations per seed; the failure is overwhelming and consistent regardless of the critic’s definitional concerns. | next: Pin down a precise definition of  $\text{FourierTransform}$  and  $\text{MinimalFourierCoefficient}$  for tropical polynomials (e.g., interpret  $f$  as a real-valued function on  $G$ , use min over nontrivial characters of  $|\hat{F}|$ ), and  $r$